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C3 — Chemicals of the Natural Environment

OCR GCSE Chemistry B (Twenty First Century Science, J258) — Chapter C3 · All papers (J258/01–04)

Content marked **(HT)** is tested only on the Higher tier papers (J258/03 and J258/04). Everything else can be tested on both tiers.

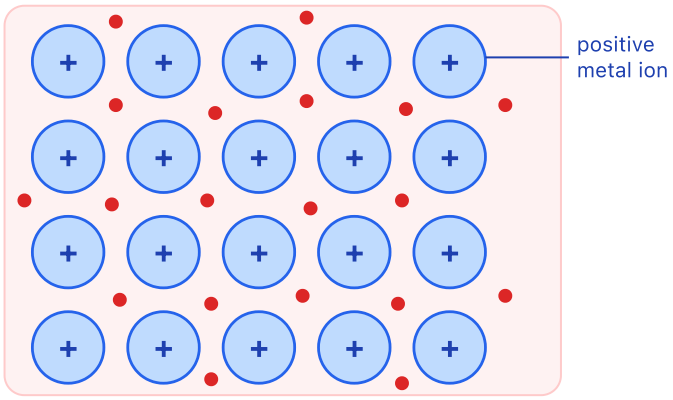
C3.1 How Are the Atoms Held Together in a Metal?

> **Metallic bonding** is the strong electrostatic attraction between a lattice of **positive metal ions** and a 'sea' of **delocalised electrons** (outer-shell electrons that are free to move through the whole structure).

In the model of metal structure, the metal atoms are packed **closely together in a regular giant structure** (a lattice). Each atom has lost its outer-shell electron(s), which become delocalised. The attraction acts in all directions, so the bonding holds the whole giant structure together — there are no separate molecules.

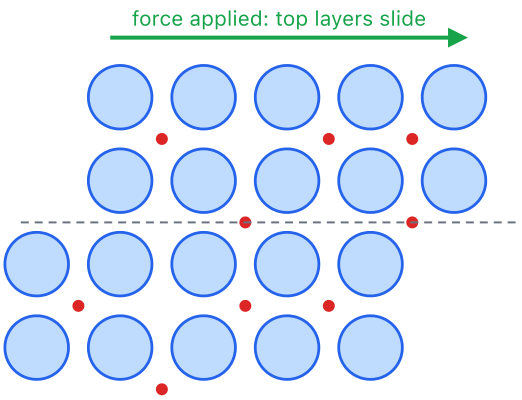
The Model of Metallic Bonding

Giant lattice of ions in a sea of electrons



delocalised electron (free to move anywhere in the lattice)

Why metals are malleable



layers of ions slide over each other but the electrons still hold them together → no shattering

Bulk Properties Explained by the Model

Property	Explanation using metallic bonding
High melting and boiling points	Strong electrostatic attraction between the positive ions and the delocalised electrons; a lot of energy is needed to overcome it.
Good electrical conductors	Delocalised electrons are free to move through the structure and carry charge when a potential difference is applied.
Good thermal conductors	Free electrons carry energy quickly through the lattice (and the closely packed ions pass on vibrations).
Malleable (can be hammered into shape) and ductile (drawn into wires)	Layers of ions slide over each other; the bonding is non-directional, so the electrons keep holding the ions together in their new positions.
Usually strong and dense	Ions are packed closely together and strongly bonded.

These properties make metals useful: copper for electrical wiring (conductor, ductile), aluminium and steel for pans (thermal conductor, high melting point), steel for car bodies (strong, malleable). Uses of metals are one of the spec's examples of science improving lives **(IaS4)**.

> **Exam Tip: Ideas about Science (IaS3) — using a model.** Chemists use this model to explain metal properties, but it has limits: real ions are not hard balls, the diagram is 2D while the lattice is 3D, electrons are not really fixed dots, and the simple model does not explain why, for example, iron has a much higher melting point than sodium. When asked to "use the model", always mention *ions*, *delocalised electrons* and the *attraction* between them.

C3.2 How Are Metals with Different Reactivities Extracted?

Finding an Order of Reactivity

Metals can be put in order of reactivity using experimental results from three types of reaction:

- **With water** — metal + water → metal hydroxide + hydrogen (with steam: metal oxide + hydrogen).
- **With dilute acid** — metal + acid → salt + hydrogen. The faster the fizzing (or the bigger the temperature rise), the more reactive the metal.
- **Displacement** — a more reactive metal displaces a less reactive metal from a solution of its compound (or from its oxide when heated).

Metal	Cold water	Dilute acid
Potassium, K	Very vigorous; melts, fizzes, hydrogen ignites (lilac flame)	Dangerously violent — not done in school
Sodium, Na	Vigorous; melts into a ball and fizzes around	
Calcium, Ca	Steady fizzing; solution goes cloudy	
Magnesium, Mg	Very slow (reacts well with steam)	Vigorous fizzing
Aluminium, Al	No visible reaction (protective oxide layer)	Slow at first, then steady
Zinc, Zn	No reaction (reacts with steam)	Steady fizzing
Iron, Fe	Very slow (rusting)	Slow fizzing
Lead, Pb	No reaction	Barely any reaction
Copper, Cu / Silver, Ag	No reaction	No reaction (below hydrogen)

Practical work: investigate the reactivity of metals with water and dilute acid (compare fizzing or temperature rise using the *same* mass/surface area of metal and the same volume and concentration of acid); and heat zinc, iron and copper with the oxides of the other two. For example, zinc powder heated with copper(II) oxide glows and leaves red-brown copper, but copper

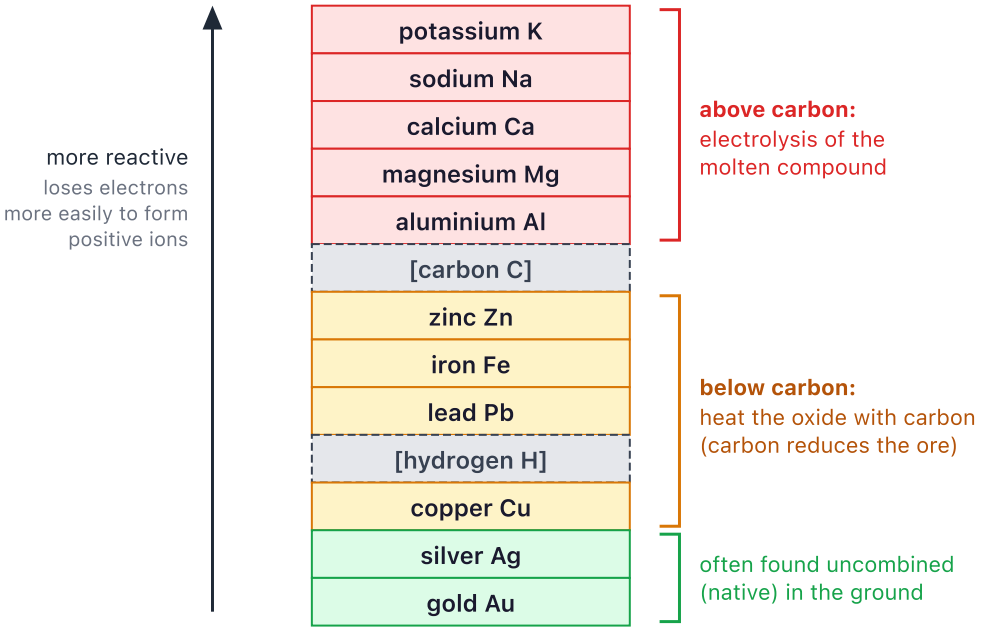
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heated with zinc oxide shows no change. These activities can contribute to **PAG1 (Reactivity trend)**.

$$\text{zinc} + \text{copper(II) oxide} \rightarrow \text{zinc oxide} + \text{copper}$$

$$\text{Zn(s)} + \text{CuO(s)} \rightarrow \text{ZnO(s)} + \text{Cu(s)}$$

The Reactivity Series and How Each Metal Is Extracted



C and H are non-metals placed for comparison

Reactivity and Forming Positive Ions

When metals react they form **ionic compounds**: the metal atoms **lose** one or more outer electrons to become **positive ions** (e.g. Na → $Na^+ + e^-$; Mg → $Mg^{2+} + 2e^-$). The more easily a metal forms its positive ion, the more reactive it is. Potassium loses its single outer electron very easily, so it reacts violently with water; copper and silver hold on to their electrons, so they do not react with water or dilute acids. Metals below **hydrogen** cannot displace hydrogen from water or acids.

Writing Formulae and Balanced Equations

Atoms are neither created nor destroyed in a reaction (**conservation of mass**), so a balanced equation has the same number of each type of atom on both sides. Balance by changing the large numbers *in front of* formulae, never the small subscripts. Include state symbols: (s), (l), (g), (aq).

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$$2Na(s) + 2H_2O(l) \rightarrow 2NaOH(aq) + H_2(g)$$
$$Ca(s) + 2H_2O(l) \rightarrow Ca(OH)_2(aq) + H_2(g)$$
$$Mg(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$$
$$Zn(s) + CuSO_4(aq) \rightarrow ZnSO_4(aq) + Cu(s)$$

Ionic Equations (HT)

(HT) An **ionic equation** shows only the ions (and atoms) that change in the reaction, leaving out **spectator ions** that stay the same. Ionic equations are useful for **displacement reactions** because they show what happens to the metal ions — the more reactive metal atoms lose electrons and the less reactive metal ions gain them.

Worked Example: (HT) Write the ionic equation for zinc reacting with copper(II) sulfate solution.
Full equation: $Zn(s) + Cu^{2+}(aq) + SO_4^{2-}(aq) \rightarrow Zn^{2+}(aq) + SO_4^{2-}(aq) + Cu(s)$
Sulfate ions are unchanged (spectators) → cross them out:
$Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$
Zinc atoms each lose 2 electrons ($Zn \rightarrow Zn^{2+} + 2e^-$); copper ions each gain 2 electrons ($Cu^{2+} + 2e^- \rightarrow Cu$). Check: charges balance (2+ on each side).

(HT) Similarly, a metal with an acid: $Mg(s) + 2H^+(aq) \rightarrow Mg^{2+}(aq) + H_2(g)$ (chloride ions are spectators).

Extracting Metals: Where Does Carbon Fit?

Most metals are found in **ores** as compounds (often oxides or sulfides). How a metal is extracted depends on its reactivity. Carbon is a non-metal but can be placed in the reactivity series **between aluminium and zinc**.

- **Metals below carbon** (zinc, iron, lead, copper) are extracted by heating the ore with carbon. Carbon displaces the metal: the metal compound is **reduced** (loses oxygen) and carbon is **oxidised** (gains oxygen).
- **Metals above carbon** (potassium to aluminium) hold oxygen more strongly than carbon does, so carbon cannot reduce them. They are extracted by **electrolysis** of the molten compound

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(C3.3), which needs large amounts of electricity and so is expensive.

> **Worked Example:** Extraction of zinc. The main ore, zinc blende, is zinc sulfide.
> 1. Roast the ore in air to make zinc oxide: $2ZnS(s) + 3O_2(g) \rightarrow 2ZnO(s) + 2SO_2(g)$
> 2. Heat the zinc oxide with carbon (coke) in a furnace: **$ZnO(s) + C(s) \rightarrow Zn(g) + CO(g)$**
> The furnace is hotter than zinc's boiling point (907 °C), so zinc forms as a vapour that is condensed and collected. Zinc oxide is reduced; carbon is oxidised. The sulfur dioxide from roasting must be captured (it causes acid rain) — it is usually turned into sulfuric acid.

Iron is extracted in the same way in a blast furnace, where carbon monoxide made from coke does the reducing: $Fe_2O_3(s) + 3CO(g) \rightarrow 2Fe(l) + 3CO_2(g)$. (Much of today's zinc is in fact made by electrolysing zinc sulfate solution, but the spec expects you to explain the carbon-reduction method.)

Biological Methods of Extraction (HT)

(HT) Rich ores are running out. Scientists are developing ways to extract the less reactive metals (such as copper) from **low-grade ores** and **waste material** (e.g. old mine spoil heaps) using living organisms.

Method (HT)	How it works
Bacterial extraction (bioleaching)	Bacteria feed on low-grade ore and produce a solution (leachate) containing metal ions such as Cu^{2+}. The metal is then recovered from the solution by displacement with scrap iron or by electrolysis.
Phytoextraction (phytomining)	Plants are grown on soil containing low concentrations of the metal; they absorb metal ions through their roots. The plants are harvested and burned; the ash contains metal compounds, from which the metal is extracted (e.g. by electrolysis or displacement).

(HT) **Evaluating** these methods — weigh the benefits against the drawbacks:

- **For:** can use low-grade ores and waste, reducing the need to mine 'new' ores (less landscape damage, less waste rock); lower energy costs than smelting; less sulfur dioxide and CO_2; removes toxic metals from land and so reduces the amount of toxic metals in landfill.

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- **Against:** very **slow** (phytoextraction takes whole growing seasons) and produces only **small quantities** of metal; bioleaching solutions can be toxic and acidic and must not escape into groundwater; the metal still has to be extracted from the leachate or ash.

> **Exam Tip: Ideas about Science (IaS4).** Metal extraction has environmental impacts — quarrying and mining scar the landscape, smelting uses lots of energy and releases CO_2 and SO_2, waste heaps can leach toxic metals. Scientists mitigate these by recycling, capturing gases and developing biological methods. In "evaluate" questions, compare the *risks, costs and benefits* of each course of action and finish with a justified judgement.

C3.3 What Are Electrolytes and What Happens During Electrolysis?

> **Electrolysis** is the decomposition of an **electrolyte** by an electric current. An **electrolyte** is an ionic compound that is **molten or dissolved in water**, so its ions are free to move.

- Solid ionic compounds are **not** electrolytes: their ions are held in fixed positions in the lattice.
- Covalent substances such as sugar solution or ethanol do not conduct — they contain no ions.
- The positive electrode is the **anode**; the negative electrode is the **cathode**. **Inert electrodes** (graphite or platinum) do not react themselves.
- Positive ions (metal ions or H^+) are attracted to the **cathode**, where they **gain electrons** to become neutral atoms. Negative non-metal ions are attracted to the **anode**, where they **lose electrons**.

Practical work: to investigate which substances are electrolytes, dip two electrodes connected in series with a cell and a bulb (or ammeter) into each substance — solid and dissolved salt, sugar solution, ethanol, molten wax — and look for current and for products at the electrodes.

> **Rule (inert electrodes):** metals (or hydrogen) are formed at the **cathode**; non-metals are formed at the **anode**.

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Molten Binary Ionic Compounds

A **binary** compound contains just two elements. When molten, the only ions present are the metal ion and the non-metal ion, so the products are simply the two elements:

Molten electrolyte	At the cathode (–)	At the anode (+)
Lead(II) bromide, $PbBr_2$	Lead (molten, silvery beads)	Bromine (brown vapour)
Sodium chloride, NaCl	Sodium	Chlorine
Zinc chloride, $ZnCl_2$	Zinc	Chlorine
Aluminium oxide, Al_2O_3	Aluminium	Oxygen

Half Equations, Oxidation and Reduction (HT)

(HT) A **half equation** shows what happens at one electrode, using e^- for electrons. Atoms and charges must both balance (conservation of mass).

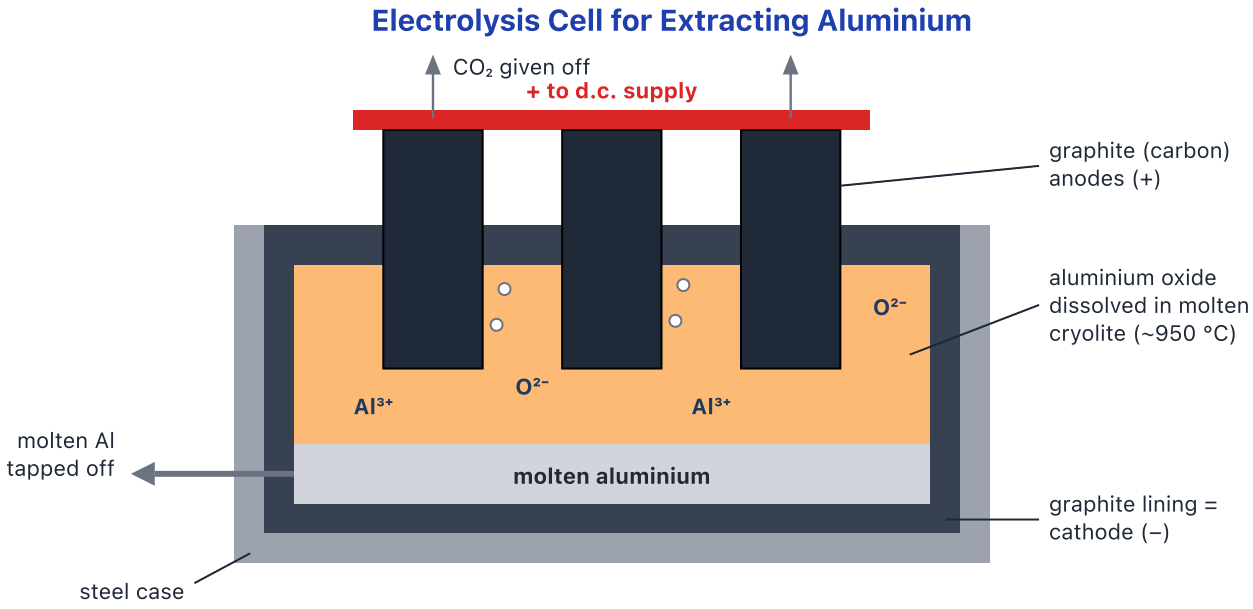
(HT) Cathode: $Pb^{2+} + 2e^- \rightarrow Pb$ Anode: $2Br^- \rightarrow Br_2 + 2e^-$

(HT) **Oxidation is loss** of electrons; **reduction is gain** of electrons ("OIL RIG"). In electrolysis, **reduction happens at the cathode** (positive ions gain electrons) and **oxidation happens at the anode** (negative ions lose electrons). So in molten lead bromide, Pb^{2+} ions are reduced and Br^- ions are oxidised.

Extracting Aluminium by Electrolysis

Aluminium is above carbon, so it is extracted by electrolysis of molten **aluminium oxide** (from the ore bauxite). The oxide melts above 2000 °C, so it is dissolved in molten **cryolite**, which lets the cell run at about 950 °C — still very hot.

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- **At the cathode:** aluminium ions gain electrons to form aluminium atoms; molten aluminium sinks and is tapped off.
- **At the anode:** oxide ions lose electrons to form oxygen molecules. The hot oxygen reacts with the carbon anodes to form carbon dioxide (C + O_2 → CO_2), so the anodes burn away and must be replaced regularly.
- The process uses a **large amount of energy**: to keep the electrolyte molten and for the electric current itself. This is why aluminium plants are often sited near cheap hydroelectric power, and why recycling aluminium is so worthwhile.

$$\text{Overall: } 2Al_2O_3(l) \rightarrow 4Al(l) + 3O_2(g)$$

$$\text{(HT) Cathode: } Al^{3+} + 3e^- \rightarrow Al \text{ (reduction)} \quad \text{Anode: } 2O^{2-} \rightarrow O_2 + 4e^- \text{ (oxidation)}$$

Electrolysis of Aqueous Solutions — Competing Reactions

Some extraction methods (for example recovering metals from waste heaps) give a **dilute aqueous solution** of metal ions. In any aqueous solution, water also provides **H^+** and **OH^-** ions, so two kinds of ion compete at each electrode:

- **Cathode (–):** the *less reactive* of the metal and hydrogen is discharged. Less reactive metals such as **copper and silver** are deposited. If the metal is more reactive than hydrogen (e.g. sodium, potassium), **hydrogen gas** forms instead.

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- **Anode (+): oxygen** usually forms (from hydroxide ions), unless the solution contains a **concentrated halide** — a concentrated chloride solution gives **chlorine**. Sulfate and nitrate ions are never discharged.

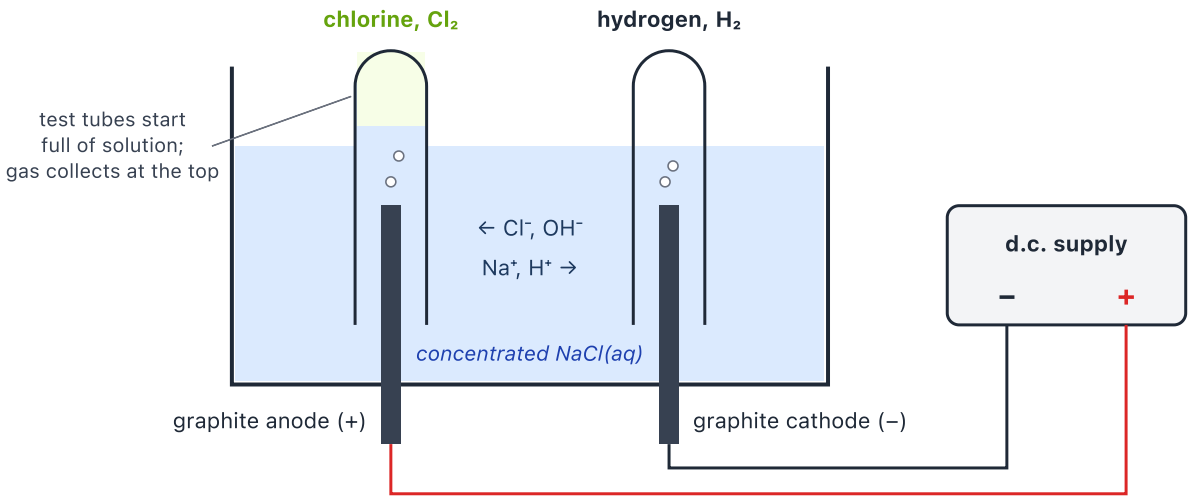
Aqueous solution (inert electrodes)	Cathode (−)	Anode (+)	Left in solution
Copper(II) sulfate, $CuSO_4$	Copper	Oxygen	Sulfuric acid
Silver nitrate, $AgNO_3$	Silver	Oxygen	Nitric acid
Concentrated sodium chloride, NaCl	Hydrogen	Chlorine	Sodium hydroxide
Very dilute sodium chloride	Hydrogen	Mostly oxygen	—
Dilute sulfuric acid, H_2SO_4	Hydrogen	Oxygen	Sulfuric acid (more concentrated)

(HT) Cathode: $2H^+ + 2e^- \rightarrow H_2$ or $Cu^{2+} + 2e^- \rightarrow Cu$

(HT) Anode: $2Cl^- \rightarrow Cl_2 + 2e^-$ or $4OH^- \rightarrow O_2 + 2H_2O + 4e^-$

Technique: Electrolysis of an Aqueous Salt Solution (PAG2)

Electrolysis of Sodium Chloride Solution, Collecting the Gases



1. Pour the salt solution into the cell; fill two small test tubes with the solution and invert one over each inert (graphite) electrode.
2. Connect the electrodes to a low-voltage d.c. supply (e.g. 4–6 V) and switch on.
3. Observe each electrode; collect the gases and test them:
   - **Hydrogen** — burns with a squeaky pop with a lighted splint.
   - **Oxygen** — relights a glowing splint.
   - **Chlorine** — turns damp blue litmus paper red and then bleaches it white.
   - A metal deposit (e.g. pink-brown copper from $CuSO_4$) shows as a coating on the cathode.

Practical work: investigate how the **concentration** of the sodium chloride solution, the **current** and the **voltage** affect the products — e.g. measure the volume of gas at each electrode in a fixed time, and test whether chlorine or oxygen forms as the solution gets more dilute. Change one variable at a time and control the others (electrode size and spacing, time, temperature). **Safety:** chlorine is toxic — use low voltages, short run times and a well-ventilated room; wear eye protection. This is **PAG2 (Electrolysis)**.

> **Exam Tip:** Name the electrodes by charge as well as name: “cathode (negative)”. A common slip is saying sodium forms at the cathode in *aqueous* sodium chloride — it doesn’t; sodium is more reactive than hydrogen, so hydrogen forms.

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C3.4 Why Is Crude Oil Important as a Source of New Materials?

Crude oil is a **mixture of hydrocarbons** (compounds of hydrogen and carbon only), formed over millions of years from the remains of tiny sea organisms. It is the main source of hydrocarbons, used both as **fuels** and as a **feedstock** (raw material) for the **petrochemical industry**.

Modern Life Depends on Hydrocarbons

Fuels from crude oil power almost all transport (petrol, diesel, kerosene for jets) and much heating and electricity generation. As a feedstock, crude oil is made into polymers (plastics and synthetic fibres), solvents, detergents, lubricants, dyes, cosmetics and medicines. Almost every consumer product uses crude oil in its manufacture or its transport.

Crude oil is a **finite (non-renewable) resource**: it formed over millions of years and is being used far faster than it forms. At the current rate it will run out, so our present use is **not sustainable**. Most crude oil is burned as fuel; only a small fraction is used as feedstock.

Exam Tip: Ideas about Science. Decisions about crude oil (IaS3 in the spec, linked to IaS4 sustainability) must balance short-term benefits (cheap fuel, jobs, convenience) against conserving a finite resource for the future — especially as the only source of feedstock for many chemicals. A strong answer considers different groups (consumers, industry, future generations) and suggests actions such as using alternative fuels for energy and keeping oil for making materials.

Fractional Distillation

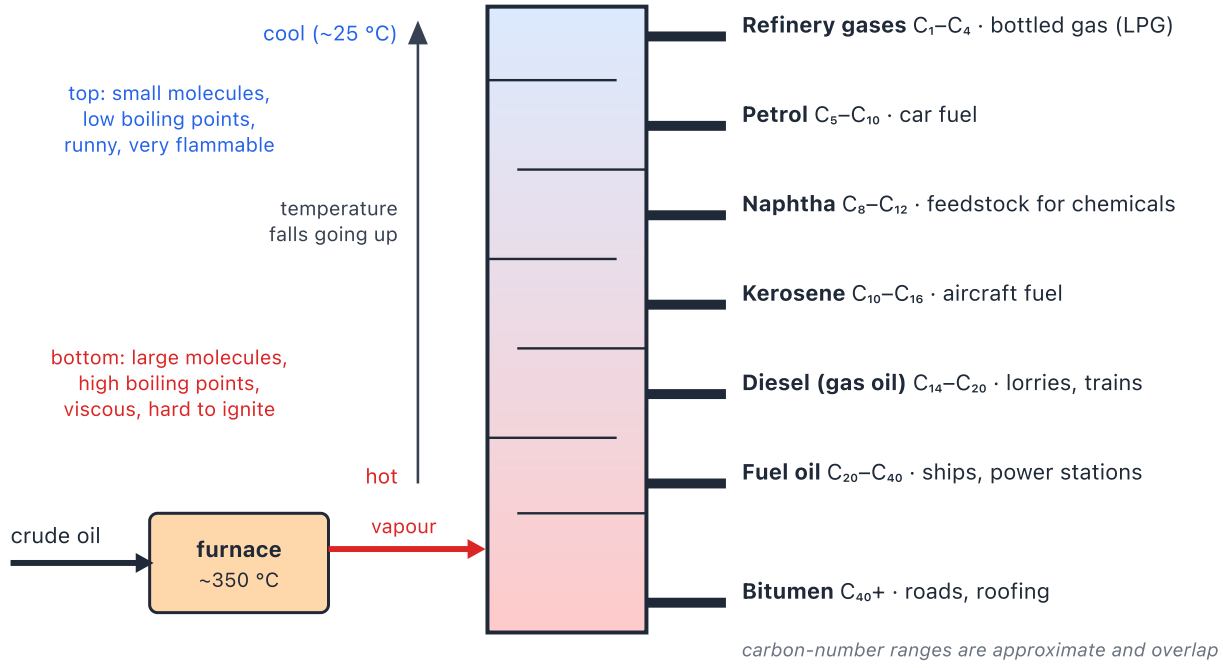
Crude oil is separated into **fractions** — groups of hydrocarbons with similar-sized molecules and a narrow range of boiling points — by **fractional distillation**. It works because hydrocarbons have different boiling points, which depend on the size of the molecules and the intermolecular forces between them.

1. Crude oil is heated in a furnace to about 350 °C so that most of it evaporates.

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2. The vapour enters near the bottom of a tall **fractionating column**, which is **hot at the bottom and cooler at the top**.
3. Vapours rise. Each hydrocarbon **condenses** when it reaches a level where the temperature is below its boiling point, and is collected on trays at that level.
4. Small molecules (low boiling points) rise to the top and leave as gases; large molecules (high boiling points) condense near the bottom. The heaviest part (bitumen) never evaporates and is drained from the base.

Fractional Distillation of Crude Oil



The fractions are still **mixtures**, mainly of **alkanes** (general formula $\mathbf{C_nH_{2n+2}}$). Going down the column, molecules get larger, so boiling point and viscosity increase while volatility and flammability decrease. In the lab, a small sample of 'synthetic crude' can be separated by fractional distillation, collecting fractions over set temperature ranges — this links to **PAG3 (Separation techniques)** and **PAG4 (Distillation)**, for which the spec suggests distilling hydrocarbons. In the lab set-up the thermometer bulb sits level with the side arm and cooling water enters the condenser at the bottom.

Alkanes: a Homologous Series

A **homologous series** is a family of compounds that have the **same general formula**, in which each member differs from the next by **CH_2**, which have **similar chemical properties** and show a **trend in physical properties**.

Alkane	Molecular formula	Boiling point / °C	State at 20 °C
Methane	CH_4	−162	gas
Ethane	C_2H_6	−89	gas
Propane	C_3H_8	−42	gas
Butane	C_4H_{10}	−1	gas

Alkanes are **saturated** (only C–C single bonds) and fairly unreactive, apart from burning.

Changes of State: Bonds versus Intermolecular Forces

Within a hydrocarbon molecule the atoms are joined by strong **covalent bonds**. Between molecules there are only weak **intermolecular forces**. When a simple molecular substance melts or boils, energy is transferred to the molecules so they move apart: **only the intermolecular forces are overcome; no covalent bonds are broken**. That takes little energy, so the melting and boiling points are low.

Bigger molecules have more contact with each other, so the total intermolecular forces are stronger; more energy is needed to separate them, so the boiling point is higher. This explains the trend in the alkanes table and the order of fractions in the column.

Molecular and Empirical Formulae

The **molecular formula** gives the actual number of atoms of each element in one molecule. The **empirical formula** is the **simplest whole-number ratio** of atoms of each element.

Compound	Molecular formula	Ratio	Empirical formula

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Ethane	C_2H_6	2 : 6 = 1 : 3	CH_3
Butane	C_4H_{10}	4 : 10 = 2 : 5	C_2H_5
Ethene	C_2H_4	2 : 4 = 1 : 2	CH_2
Methane	CH_4	1 : 4 (already simplest)	CH_4

Worked Example: A model of a hydrocarbon shows 6 carbon atoms and 14 hydrogen atoms. Give its molecular and empirical formulae.
Count from the model: molecular formula C_6H_{14}. Divide by the highest common factor, 2: 6 : 14 = 3 : 7.
Empirical formula = **C_3H_7**. (Going the other way: an alkane with empirical formula C_2H_5 and 4 carbons must be C_4H_{10}.)

Worked Example: (Illustrative data.) A sample of a hydrocarbon contains 0.40 mol of carbon atoms and 1.00 mol of hydrogen atoms. Find the empirical formula.
Ratio C : H = 0.40 : 1.00. Divide by the smaller: 1 : 2.5. Multiply by 2 to get whole numbers: 2 : 5. Empirical formula = **C_2H_5**.

Covalent Bonding in Simple Molecules

A **covalent bond** is a strong bond between two atoms formed by a **shared pair of electrons**. The two nuclei are both attracted to the shared pair. Covalent bonds form between **non-metal atoms**; sharing lets each atom reach a full outer shell.

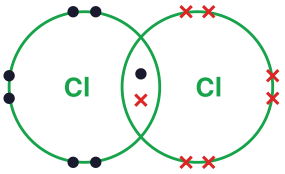
A **simple molecule** contains a small, fixed number of atoms joined by covalent bonds (e.g. H_2O, CO_2, CH_4, Cl_2, and the small molecules met in C1 such as O_2, N_2, SO_2). Each bond is **directional** — it joins two particular atoms — so molecules have definite shapes.

Dot and Cross Diagrams

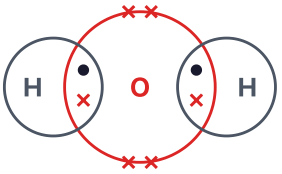
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Draw the outer shells as overlapping circles; show one atom's electrons as dots and the other's as crosses; put each shared pair in the overlap. Hydrogen ends up with 2 electrons; the other atoms with 8.

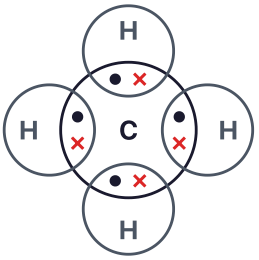
Dot and Cross Diagrams (outer shells only)



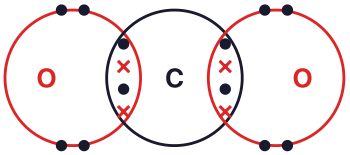
chlorine, Cl_2
1 shared pair; 3 lone pairs on each Cl



water, H_2O
2 shared pairs; 2 lone pairs on O



methane, CH_4
4 shared pairs; no lone pairs



carbon dioxide, CO_2 (O=C=O)
two double bonds (2 shared pairs each); 2 lone pairs on each O

● electrons from one atom
× electrons from the other atom
pair in an overlap = 1 covalent bond

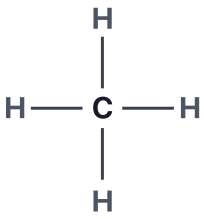
Representing Molecules in 2D and 3D

Molecules can be shown as a **molecular formula** (C_2H_6), an **empirical formula** (CH_3), a **displayed formula** (every atom and every bond drawn as a line), a **dot and cross diagram**, or a 3D **ball and stick** or **space-filling** model. You need to translate between them, e.g. count atoms in a model to write a formula, or turn a 3D model into a flat displayed formula.

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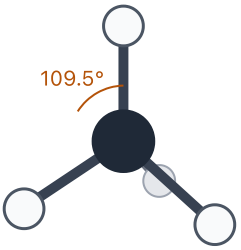
Three Ways to Represent Methane, CH_4

Displayed formula (2D)



shows every bond clearly
but drawn flat with 90° angles;
real shape is not flat

Ball and stick (3D)



shows the tetrahedral 3D shape
but bonds are not sticks and atoms
are not really separated by gaps

Space-filling (3D)



shows relative sizes and how
atoms overlap
but hides the bonds and angles

the fourth (pale) H sits behind the C atom in the 3D models

Representation	Useful because...	Limitations
Dot and cross	Shows where every outer electron is, which atom it came from, shared and lone pairs	Electrons are actually identical; doesn't show shape or bond angles; electrons are not fixed in position
Displayed formula	Shows every atom and bond; easy to draw and count	Flat (2D) — wrong angles; no lone pairs; no relative atom sizes
Ball and stick	Shows 3D shape and bond angles	Atoms are not hard balls; bonds are not sticks; gaps and sizes not to scale; no electrons shown
Space-filling	Closest picture of the molecule's real shape and size	Bonds and angles hidden; hard to see atoms at the back

Exam Tip: Ideas about Science (IaS3). No model is 'right'; each is chosen for what it shows. None of these models shows the intermolecular forces between molecules, which are what decide melting and boiling points.

Properties of Simple Molecular Substances

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- **Low melting and boiling points** — only weak intermolecular forces need to be overcome; the strong covalent bonds inside the molecules are not broken. Many are gases or liquids at room temperature.
- **Do not conduct electricity** — the molecules have no overall charge and there are no free electrons or ions.
- Larger molecules → stronger intermolecular forces → higher melting and boiling points.

Cracking

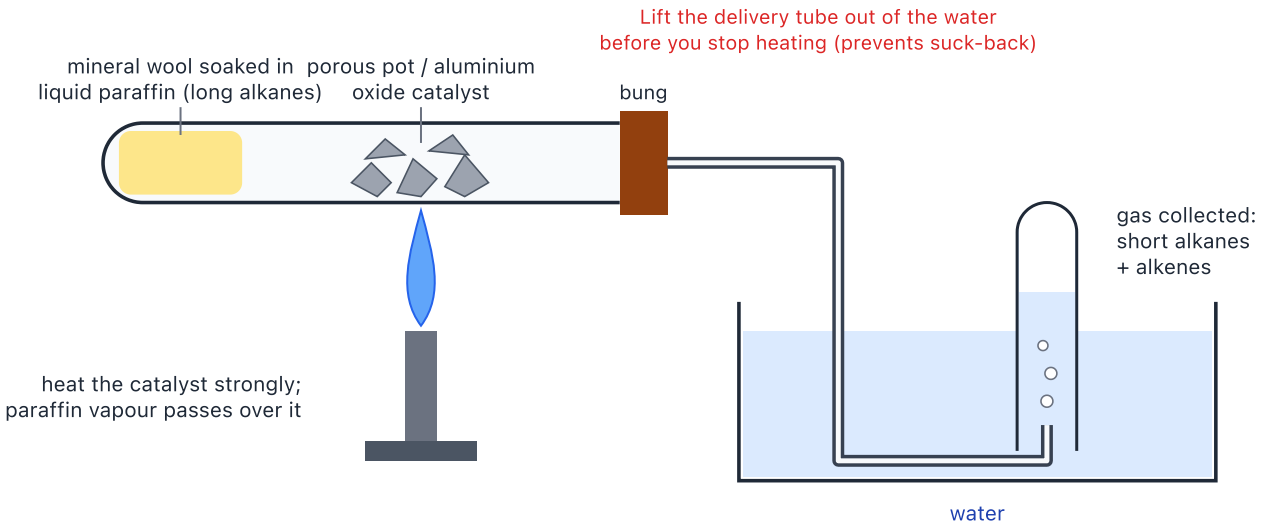
Fractional distillation gives more of the long-chain fractions than are needed, and not enough of the short-chain ones such as petrol, which are in great demand as fuels. **Cracking** breaks long-chain alkanes into **shorter, more useful alkanes** and **alkenes**.

- The long-chain alkanes are heated to vaporise them and passed over a **hot catalyst** (e.g. aluminium oxide or silica), or mixed with steam at a very high temperature.
- **Alkenes** are hydrocarbons with a **C=C double bond**. They are much more reactive than alkanes and are the starting materials for polymers and many other chemicals.

$$\text{decane} \rightarrow \text{octane} + \text{ethene}$$

$$C_{10}H_{22} \rightarrow C_8H_{18} + C_2H_4$$

Cracking Liquid Paraffin in the Laboratory



Test the gas with **bromine water**: alkenes turn it from **orange to colourless**; alkanes leave it orange.

> **Exam Tip: Ideas about Science (IaS4).** Cracking is a positive application of science: by matching supply to demand it means less crude oil has to be extracted for the same amount of petrol, so it helps conserve oil reserves and avoids wasting long-chain alkanes that are less useful.

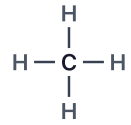
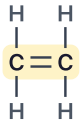
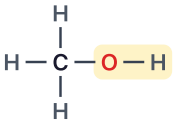
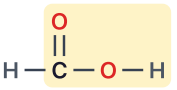
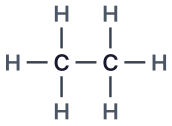
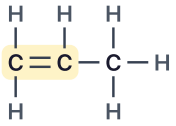
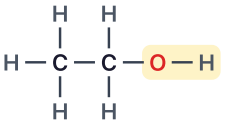
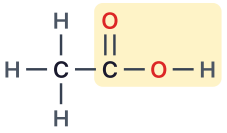
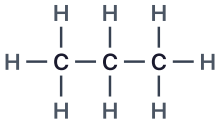
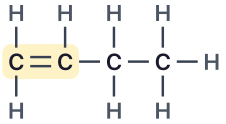
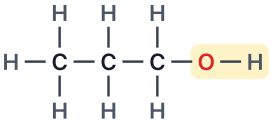
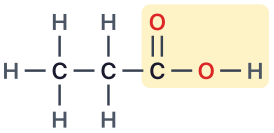
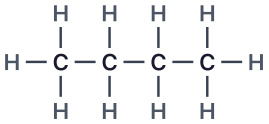
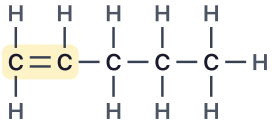
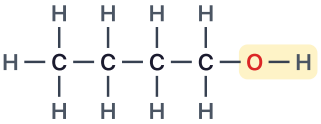
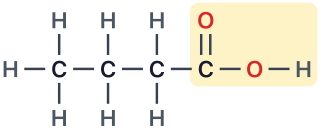
Functional Groups and Homologous Series

> A **functional group** is the atom or group of atoms that gives a homologous series its characteristic reactions. Compounds with the same functional group react in the same way.

Series	Functional group	General formula	Key properties and uses
Alkanes	none (C–C and C–H single bonds only)	C_nH_{2n+2}	Unreactive apart from burning; fuels
Alkenes	C=C double bond	C_nH_{2n}	Addition reactions across the double bond; decolourise bromine water; make polymers
Alcohols	–OH (one H of an alkane replaced by OH)	$C_nH_{2n+1}OH$	Burn as fuels; solvents that are **miscible** with water; oxidised to carboxylic acids
Carboxylic acids	–COOH	$C_nH_{2n+1}COOH$	Weak acids (pH about 3–4); react with carbonates to give CO_2; in foods (vinegar is ethanoic acid, citric acid) and personal care products

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Displayed Formulae of the First Four Members of Each Homologous Series

Alkanes C_nH_{2n+2} · no functional group	Alkenes C_nH_{2n} · C=C double bond	Alcohols $C_nH_{2n+1}OH$ · –OH group	Carboxylic acids –COOH group
 **methane** CH_4	 **ethene** C_2H_4	 **methanol** CH_3OH	 **methanoic acid** HCOOH
 **ethane** C_2H_6	 **propene** C_3H_6	 **ethanol** C_2H_5OH	 **ethanoic acid** CH_3COOH
 **propane** C_3H_8	 **but-1-ene** C_4H_8	 **propan-1-ol** C_3H_7OH	 **propanoic acid** C_2H_5COOH
 **butane** C_4H_{10}	 **pent-1-ene** C_5H_{10}	 **butan-1-ol** C_4H_9OH	 **butanoic acid** C_3H_7COOH

Exam Tip: Numbering matters for longer chains: but-1-ene has the C=C between carbons 1 and 2; propan-1-ol has the –OH on carbon 1. In a carboxylic acid, the C of the –COOH group counts in the name (ethanoic acid has 2 carbons). When drawing, check every C has exactly **4** bonds, every H 1 and every O 2.

Reactions of Alkanes, Alkenes and Alcohols

Combustion. Alkanes, alkenes and alcohols burn in plenty of air (complete combustion) to give **carbon dioxide and water**:

$$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$$

$$C_2H_4 + 3O_2 \rightarrow 2CO_2 + 2H_2O$$

$$C_2H_5OH + 3O_2 \rightarrow 2CO_2 + 3H_2O$$

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Addition across the double bond (alkenes). The C=C opens and a molecule adds on, one atom to each carbon. There is only one product.

ethene + bromine → 1,2-dibromoethane $C_2H_4 + Br_2 \rightarrow C_2H_4Br_2$ (bromine water decolourised)

ethene + hydrogen → ethane $C_2H_4 + H_2 \rightarrow C_2H_6$ (nickel catalyst)

ethene + steam → ethanol $C_2H_4 + H_2O \rightarrow C_2H_5OH$ (acid catalyst)

Alkenes also add to each other to make **addition polymers** (C4.2).

Oxidation of alcohols to carboxylic acids. An oxidising agent (e.g. acidified potassium dichromate, or microbes in air when wine turns to vinegar) converts an alcohol into the carboxylic acid with the same number of carbon atoms:

ethanol + oxygen → ethanoic acid + water

$C_2H_5OH + O_2 \rightarrow CH_3COOH + H_2O$

Predicting Products: the Generality of Functional Group Reactions

It is the **functional group** that decides how an organic compound reacts, and every member of a homologous series has the same functional group. So once you know one reaction, you can predict it for any member — including ones you have never met:

Worked Example: Predict the products for: (a) pentane burning completely; (b) propene + bromine; (c) oxidising butan-1-ol.
(a) $C_5H_{12} + 8O_2 \rightarrow 5CO_2 + 6H_2O$ (check O: 16 = 10 + 6)
(b) $C_3H_6 + Br_2 \rightarrow C_3H_6Br_2$ (1,2-dibromopropane: a Br on each carbon of the former C=C)
(c) $C_4H_9OH \rightarrow C_3H_7COOH$, **butanoic acid** (same number of carbons; the end CH_2OH becomes COOH)

Practical work: investigate the reactions of alkanes, alkenes and alcohols — e.g. compare how they burn, shake each with bromine water, and test alcohols and their oxidation products with universal indicator.

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Exam Tip: **Ideas about Science (IaS3).** Chemists use models of functional groups to represent and predict the reactions of whole homologous series. A model like this is powerful because it lets you make predictions you can test.

C3 Summary & Key Equations

- Metallic bonding: positive ions + delocalised electrons → conductors, malleable, high melting point
- Reactivity: K Na Ca Mg Al [C] Zn Fe Pb [H] Cu Ag Au — more reactive = forms positive ions more easily
- Below carbon → reduce with carbon ($ZnO + C \rightarrow Zn + CO$); above carbon → electrolysis of molten compound
- Electrolysis: cations to cathode (gain e^-, reduced); anions to anode (lose e^-, oxidised)
- Aluminium: Al_2O_3 in molten cryolite; Al at cathode; O_2 at anode burns carbon anodes to CO_2
- Aqueous: less reactive of metal/H at cathode; O_2 at anode unless concentrated halide (Cl_2)
- Crude oil: finite mixture of hydrocarbons; fractional distillation by boiling point; alkanes C_nH_{2n+2}
- Boiling only overcomes intermolecular forces, never covalent bonds
- Cracking: long alkane → shorter alkane + alkene; alkenes decolourise bromine water
- Functional groups: C=C, –OH, –COOH decide reactions for the whole series

Exam Tip: In “explain the boiling point” questions, the words *intermolecular forces* must appear. Saying “covalent bonds break when it boils” loses the mark every time.

Specification Checklist

Ref	I can...	Tier
C3.1.1	describe the nature and arrangement of bonds in metals	F + H

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C3.1.2	explain bulk properties of metals from metallic bonding	F + H
C3.2.1	deduce a reactivity order from water, acid and displacement results	F + H
C3.2.2	relate reactivity (K to Ag, with H) to forming positive ions	F + H
C3.2.3	write formulae and balanced equations; ionic equations are (HT)	F + H; ionic (HT)
C3.2.4	explain extraction using carbon's position, including zinc	F + H
C3.2.5	explain why electrolysis is used for some metals	F + H
C3.2.6	evaluate bacterial and phytoextraction	(HT)
C3.3.1	describe electrolysis in terms of ions and electrode reactions	F + H
C3.3.2	predict products of molten binary ionic compounds	F + H
C3.3.3	recall metals/hydrogen at cathode, non-metals at anode	F + H
C3.3.4	write half equations	(HT)
C3.3.5	explain oxidation and reduction as electron loss and gain	(HT)
C3.3.6	explain extraction of aluminium by electrolysis	F + H
C3.3.7	describe competing reactions in aqueous electrolysis	F + H
C3.3.8	describe the technique of electrolysing a salt solution (PAG2)	F + H
C3.4.1–2	recall crude oil as source and feedstock; dependence; finite	F + H
C3.4.3	describe and explain fractional distillation (PAG3)	F + H
C3.4.4	describe fractions as mainly alkanes C_nH_{2n+2}	F + H
C3.4.5	explain changes of state using bonds versus intermolecular forces	F + H

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C3.4.6–7	deduce empirical formulae using ratio (M1c)	F + H
C3.4.8–10	describe and explain covalent bonding; draw dot and cross diagrams	F + H
C3.4.11–13	convert between 2D and 3D, diagrams and numbers; limitations of models (M5b, M4a)	F + H
C3.4.14	explain properties of simple molecules	F + H
C3.4.15	describe cracking	F + H
C3.4.16–17	recognise functional groups; name and draw first four alkanes, alkenes, alcohols, carboxylic acids	F + H
C3.4.18–19	predict products of combustion, addition and oxidation; generality of functional group reactions	F + H

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